Due on **Friday, November 6, 2026**; It can be turned in online via Canvas.

1 Objective

Future Artificial Intelligence, Ltd. (FAIL) were very happy with the job that the VT team did on the DFF and hence they are eager to continue working with us. They noticed that there is a need for a few additional important Boolean functions, the Majority Gate and the One Hot Gate:

Majority Gate				One Hot			
A	B	C	Y	A	B	C	Y
0	0	0	0	0	0	0	0
0	0	1	0	0	0	1	1
0	1	0	0	0	1	0	1
0	1	1	1	0	1	1	0
1	0	0	0	1	0	0	1
1	0	1	1	1	0	1	0
1	1	0	1	1	1	0	0
1	1	1	1	1	1	1	0

You must design the schematic, layout, abstract and symbol view for these critical cells so that they fit into the 9T standard cell library. You may examine the library, copy cells and stretch existing cells to fit your needs.

As part of this, you will also need to create a few entries that will later be used for synthesis and automatic placement and routing.

2 Requirements

1. You will submit a report that must include the following sections:
2. General Description, Layout with measurement details, dynamic power, static power, delay (low-to-high, high-to-low, etc.), the area of your cell boundary. You must test your cells thoroughly (e.g., what happens when the different combinations of input change high-to-low, low-to-high, etc.). You must use Spectre.
3. In addition, you must include an extracted netlist.
4. On your cover page, you must make a summary table with all of simulation results from bullet 2 for each designed cell.

3 Tools

You will use Virtuoso and Spectre.

4 Grading

A report that includes all requirements from Section 2 will receive full credit. In addition, the team that has the lowest area and dynamic power (worst case transition in the TT corner) will receive a 5% bonus. One bonus will be available to 4540 students and one bonus will be available to 5545

students. To receive consideration for this credit, it must be included in the summary table on the cover page (no exceptions). A minimum of 5% points will be deducted for each missing section in your report. There is no length requirement, so you can be brief in your writing, but the writing must be a result of your team's efforts. *A reminder that plagiarism is a violation of the Honor Code and will result in an automatic grade of 0 on the assignment. Please include references to any work that is not your own and never directly copy anyone else's work.*